

RiskPulse AI

Real-Time AI Financial Risk Intelligence &
Event-Driven Portfolio Stress Testing Platform

S&P Global & CRISIL Campus Hackathon 2026

Candidate: Shatadru Adhikary

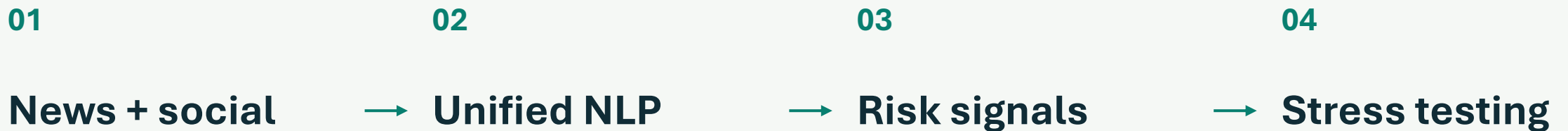
College / campus: Vellore Institute of Technology

College email: shatadru.23bce8160@vitapstudent.ac.in

Problem & approach

External information moves faster than a static portfolio view.

Risk teams need to see which event matters, why it matters, and how an illustrative scenario affects their exposures.

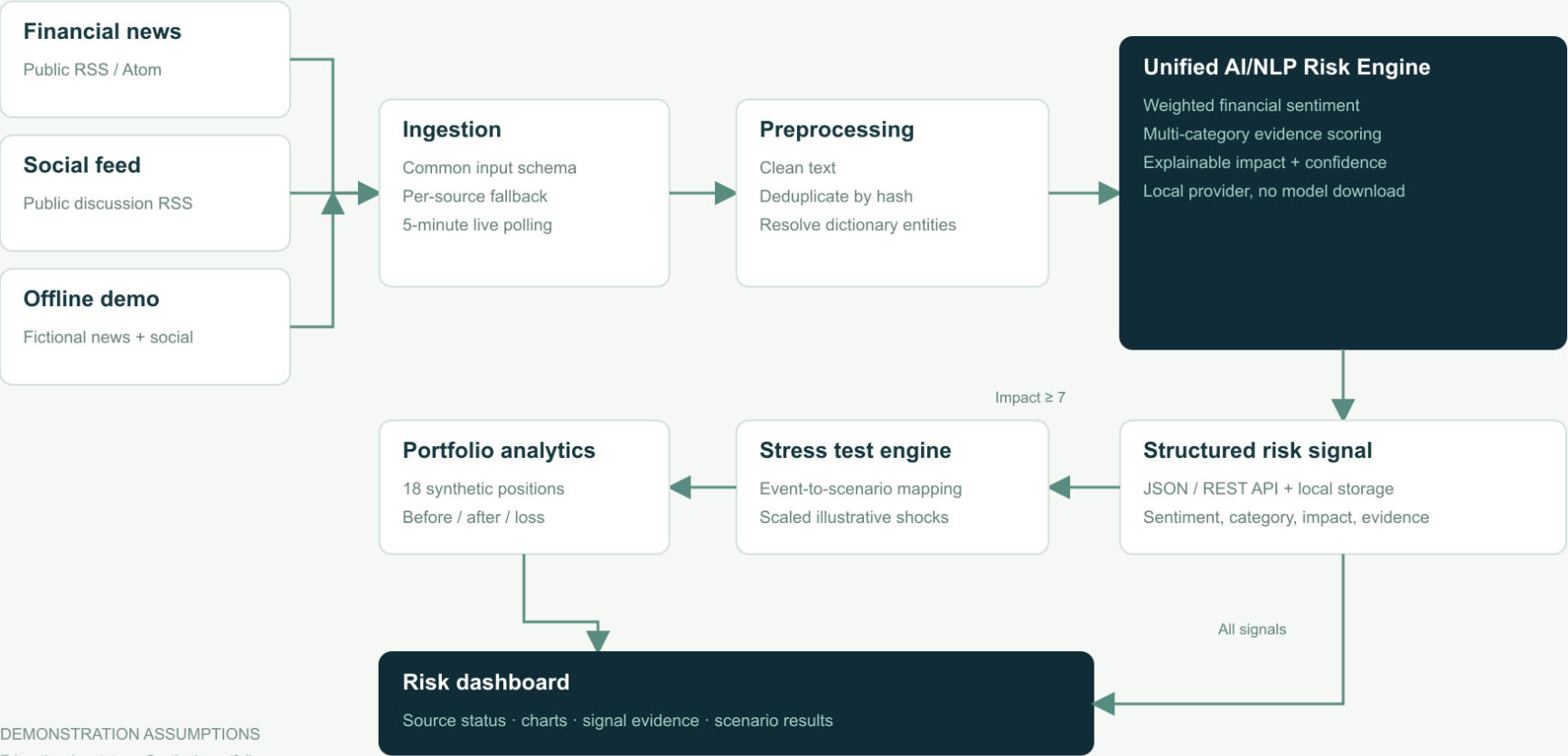


Decision support starts with traceable evidence and a visible link to portfolio impact.

System architecture

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Unified AI/NLP Risk Engine

Sentiment

Weighted financial phrases with negation and intensifiers. Range -1 to +1.

Classification

Eleven event classes compete on weighted evidence. Confidence is heuristic.

Impact

Severity, sentiment, urgency, systemic context, entities, source and recency.

Entities & evidence

Configurable names and tickers. Matched terms and every impact factor remain visible.

Geopolitical demo: sentiment -0.982 / impact 9.5/10

Event-driven portfolio stress testing

High-impact event → scenario → shocks → valuation → portfolio impact

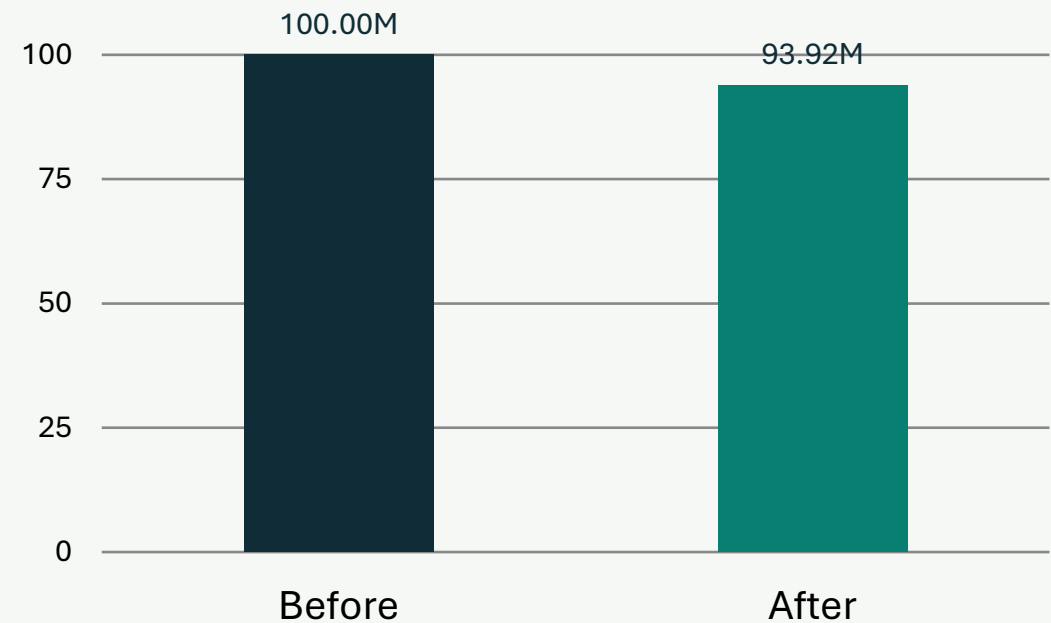
Geopolitical event 9.5/10

Equity -9.39%

Rates +94 bp

Credit spreads +141 bp

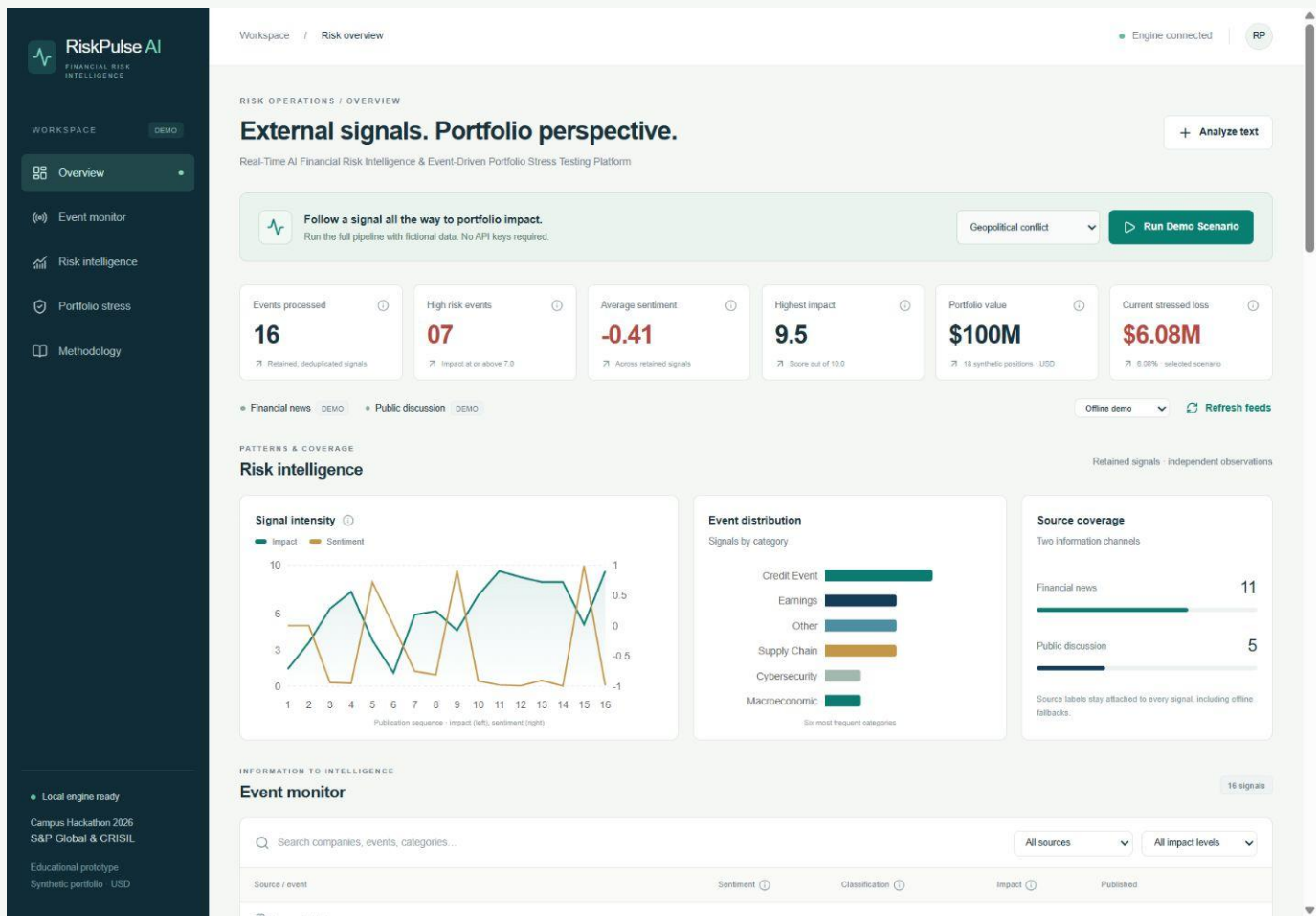
FX -4.69%



\$6.08M illustrative loss / 6.08% drawdown

DEMONSTRATION ASSUMPTIONS. Independent scenario on a \$100M synthetic portfolio.

Results & domain impact



5

reproducible demo scenarios

18

synthetic positions across six asset classes

One evidence trail links a financial headline to a scenario and asset-level losses.

Limitations & future work

Current boundaries

Rule-based English NLP can miss context.

Public feeds can fail or contain rumors.

Shocks are illustrative and uncalibrated.

Local persistence supports one server process.

Educational prototype. No guaranteed predictions or investment advice.

Next improvements

Validate finance-specific transformer models.

Add source corroboration and entity resolution.

Calibrate scenarios to historical market data.

Extend to VaR, expected shortfall and contagion.